

Unit 7.2 Acid Base Multiple Choice Practice

1. Nitrous acid, HNO_2 , has a pK_a value of 3.3. If a solution of nitrous acid is found to have a pH of 4.2, what can be said about the concentration of the conjugate acid/base pair found in solution?

- a. $[\text{HNO}_2] > [\text{NO}_2^-]$ c. $[\text{H}_2\text{NO}_2^+] > [\text{HNO}_2]$
 b. $[\text{NO}_2^-] > [\text{HNO}_2]$ d. $[\text{HNO}_2] > [\text{H}_2\text{NO}_2^+]$

2. A buffer solution can be formed by dissolving equal moles of:

- a. HF and NaOH c. CH_3COOH and NaCl
 b. KBr and Na_3PO_4 d. HF and NaF

3. Which of the following changes would affect the pH of a buffer solution?

- ~~I~~ Doubling the amount of acid and conjugate base used.
~~II~~ Doubling the amount of water in the solution.
 III. Adding a small amount of strong acid or strong base.

← same $[\text{A}^-]/[\text{HA}]$ ratio
 ✓ shifts ratio in EQ rxn!

- a. III only b. I and II only c. II and III only d. I, II, and III

4. Which of the following 1.0 M solutions would have a pH greater than 7.00?

- a. HCN *acidic* b. KNO_3 *Neutral* c. NH_4Cl *acidic* d. $\text{NaC}_2\text{H}_3\text{O}_2$ *basic*

5. A buffer solution may contain equal moles of

- a. a weak acid with its conjugate base ✓
 b. a strong acid with its conjugate base
 c. a weak acid and a strong base (yes if 1:0.5!)
 d. a strong acid and a strong base

6. What is the conjugate acid and the conjugate base of HPO_4^{2-} ?

Conjugate Acid Conjugate Base

- a. PO_4^{3-} H_2PO_4^-
 b. H_2PO_4^- PO_4^{3-}
 c. H_2PO_4^- H_3PO_4
 d. H_3PO_4 PO_4^{3-}

7. When comparing 1.0 M solutions of bases, the base with the lowest $[\text{OH}^-]$ is the

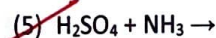
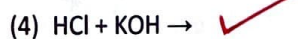
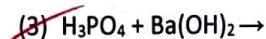
- a. weakest base and it has the largest K_b value.
- b. strongest base and it has the largest K_b value.
- c. weakest base and it has the smallest K_b value
- d. strongest base and it has the smallest K_b value.

8. Which of the following pairs of substances would make a good buffer solution?

- a. H_2SO_4 and LiOH
- b. HCl and KCl
- c. $\text{HC}_2\text{H}_3\text{O}_2$ and $\text{NaC}_2\text{H}_3\text{O}_2$
- d. HF and NaH_3

if correct ratio, could work! $\text{SO}_4^{2-} + \text{HSO}_4^-$

9. Consider the neutralization reactions between the following acid-base pairs in dilute aqueous solutions:



strong - strong only!

For which of the reactions above is the net ionic equation $\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$?

- a. 1, 3
- b. 2, 4, 5
- c. 2, 4
- d. 3, 5

10. What volume of 0.300 M KOH would be needed to completely neutralize 60.0 mL of 0.200 M H_2SO_4 ?

- a. 40.0 mL
- b. 60.0 mL
- c. 80.0 mL
- d. 100. mL

2H⁺ / 1 H₂SO₄

$$[\text{H}^+] = 2 \times 0.200 \times 0.060 = 0.024 \text{ mol OH}^-$$

$$n = 0.024 \text{ mol OH}^-$$

$$\text{H}^+ = \text{mol OH}^-$$

$$[\text{OH}^-] = [\text{NaOH}] = 0.300 = \frac{n}{V}$$

$$0.300 = \frac{0.024}{V} \quad V = 0.08 \text{ L}$$

diprotic! No MVA = MBVB

11. Which of the following indicators would be most suitable for the titration of 0.10 M lactic acid ($\text{pK}_a = 3.08$) with 0.10 M KOH(aq)?

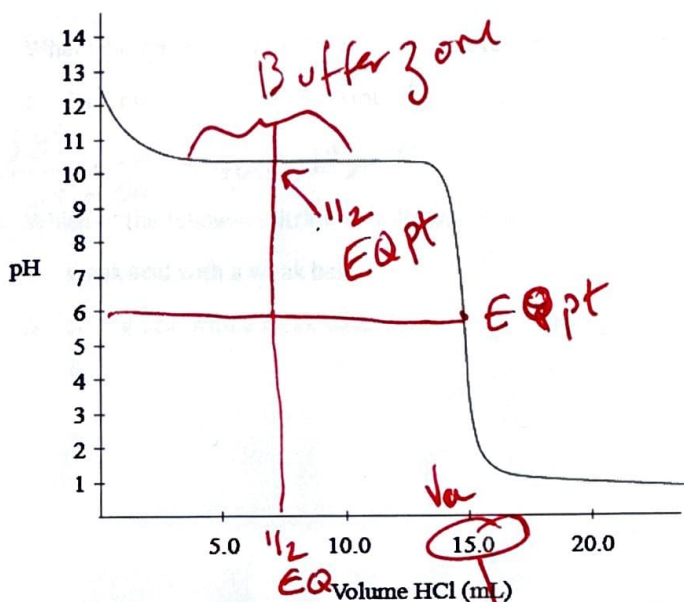
- a. phenol red ($\text{pK}_a = 6.9$)
- b. alizarin red ($\text{pK}_a = 4.5$)
- c. thymol blue ($\text{pK}_a = 1.7$)
- d. methyl orange ($\text{pK}_a = 3.4$)

weak acid! means EQ pt is BASIC

this isn't great, but it is the closest to pH > 7

Use the following information to answer questions 12–15.

A student titrates some 1.0 M HCl into 20.0 mL of methylamine (CH_3NH_2) a weak base which only accepts a single proton. The following titration curve results:



12. What is the concentration of the methylamine? $\leftarrow M_B = ?$
- a. 0.50 M b. 0.75 M c. 1.0 M d. 1.25 M

$$M_A V_A = M_B V_B \quad 1.0 \cdot 15.0 \text{ mL} = M_B \cdot 20.0$$

13. What is the approximate pK_b for methylamine? $\leftarrow$
- a. 3.5 b. 5.5 c. 10.5 d. 12.5

$$pK_b = pOH @ \frac{1}{2} EQ! \quad pH = 10.5 \quad pOH = 14 - 10.5 = 3.5 = pK_b$$

14. What buffer region of this titration is located:

- a. Below 3.0 mL c. Between 14.0 mL and 16.0 mL
- b. Between 3.0 mL and 14.0 mL d. Above 16.0 mL

15. The methylamine is replaced by 20.0 mL of sodium hydroxide of an identical concentration. If the sodium hydroxide is titrated with the 1.0 M HCl, which of the following options accurately describes the pH levels at various points during the titration when compared to the pH levels at the same point in the HCl/methylamine titration?

- | Initial pH | Equivalence pH | Ending pH |
|------------|----------------|-----------|
| a. lower | same | higher |
| b. higher | higher | same |
| c. same | higher | same |
| d. higher | lower | lower |

More basic!
EQ pt = 7

Unit 7.2 Acid Base Multiple Choice Practice

16. What volume of 0.25 M HClO_4 would be needed to completely neutralize 30.0 mL of 0.20 M Ca(OH)_2 ? ↖ 2OH⁻!

- a. 12 mL b. 24 mL c. 36 mL d. 48 mL

$$\frac{2\text{OH}^-}{1\text{Ca(OH)}_2} \times \left(0.20 \frac{\text{mol}}{\text{L}}\right) \times 0.030 \text{ L} = 0.012 \text{ mol OH}^- = \text{mol H}^+$$

$$M = n/V \quad 0.25 = 0.012/V \quad V = 0.0482 = 48 \text{ mL}$$

17. What volume of 0.100 M NaOH is required to completely neutralize 15.00 mL of 0.100 M H_3PO_4 ? ↖ 3H⁺!

- a. 5.00 mL b. 15.00 mL c. 30.00 mL d. 45.00 mL

$$\frac{3\text{H}^+}{1\text{H}_3\text{PO}_4} \times (0.100 \text{ M}) \times 0.015 \text{ L} = 0.0045 \text{ mol H}^+ = \text{mol OH}^-$$

$$M = \frac{n}{V} \quad 0.100 = \frac{0.0045}{V} \quad V = 0.045 = 45 \text{ mL}$$

18. Which of the following titrations will always have an equivalence point at a pH > 7.00?

- a. weak acid with a weak base b. strong acid with a weak base
c. weak acid with a strong base d. strong acid with a strong base

19. Which of the following changes would affect the pH of a buffer solution?

- i. adding the amount of acid and conjugate base used
ii. doubling the amount of water in the solution
iii. adding a small amount of strong acid or strong base

a. i and ii only

b. i, ii, and iii

20. A buffer solution has a pH of 7.00.

a. NaCH_3COO and CH_3COOH b. NaOH and HCl

c. weak acid and a strong base (yes if 150.51)
d. strong acid and a strong base